

SAHILSINGH KHALSA

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SUMMARY

Software Development Engineer with experience building scalable backend services, distributed workflows, full-stack applications, and production-style AI systems using Python, TypeScript, C++, AWS, SQL/NoSQL databases, and CI/CD. Built REST APIs, microservices, AI workflows, RAG pipelines, LangGraph agents, monitoring, debugging, and reliable systems.

EDUCATION

Illinois Institute of Technology

Aug 2024 - May 2026

Master of Science, Computer Science (GPA: 3.7 / 4.0)

- **Coursework:** Data Structures & Algorithms, Operating Systems, Computer Networks, Database Organization, HealthcareAI

Vidush Somany Institute of Technology and Research

Aug 2020 - May 2024

Bachelor of Science, Computer Engineering (GPA: 4.0 / 4.0)

- **Coursework:** Information Security, Software Engineering, Big Data Technologies, Probability & Statistics, Database Systems

EXPERIENCE

Software Development Engineer | Find Me LLC

Jan 2026 - May 2026

- Scaled FyndMe.net to support **9** portfolio types and **4** **template** tiers by architecting a **3-service** microservices platform across **Next.js, Node.js/Express, and FastAPI**.
- Collaborated with product and engineering teams to ship **11+** public and protected workflows by translating portfolio requirements into React/Next.js pages, API routing, and JWT-secured client-to-backend communication.
- Unified **9** **portfolio schemas** into a single **MongoDB** model using **Mongoose discriminators**, with per-type validation, visibility controls, and soft-delete support.
- Developed a containerized **AI-powered FastAPI** image-processing service using U2Net, MediaPipe, and OpenCV.

Backend Engineer | Gabriel AI

Jul 2025 - Dec 2025

- Owned campaign workflow reliability by designing **Firestore state tracking** for CSV imports, delivery updates, analytics, and **4-stage** execution across **AI voice campaigns**.
- Engineered an **AI voice-cloning pipeline** that generated **1,000+** personalized messages per campaign in under **10 minutes**, saving **33+ hours** of manual recording, by integrating ElevenLabs, AWS S3, and Drop Cowboy.
- Developed **10+ protected** campaign, billing, analytics, and voice workflows for a **0-to-1 SaaS platform** by integrating React 18, TypeScript, and Django REST.
- Collaborated across teams to secure **1M+ generated audio assets** by implementing Stripe access checks, HTTPOnly JWT auth and **CloudWatch**-backed debugging.

Software Engineer | The Special Character

Dec 2023 - May 2024

- Led backend development for advertiser-facing workflows using **TypeScript/Node.js**, building **7+ REST APIs** that enabled **4** business-critical product flows across the platform.
- Reduced noisy production failures by **32% across** critical endpoints by hardening request validation, standardizing API contracts, and adding defensive error handling to backend services.
- Improved deployment reliability by embedding automated tests into the **CI/CD pipeline**, enabling **zero-downtime** releases across **6 production deployments** and increasing confidence in rollout stability.

PROJECTS

SIGMA (Multi-Agent AI Trading System) - [Link](#)

Mar 2026 - April 2026

- Engineered a production-style **LLM agent workflow** with **6 tools** and a **6-call budget cap**, by designing a **LangGraph** decision graph that kept **LLM's reasoning** structured, auditable, and bounded at **~\$0.02 per trade decision**.
- Improved **AI decision** safety and traceability, by pairing a **Claude strategy agent** with an 8-rule deterministic risk validator that separated model reasoning from non-negotiable compliance checks for every trade.
- Built a real-time backend streaming agent decisions with **sub-100ms latency**, using a **FastAPI + PostgreSQL + WebSocket** pipeline with **async broadcasting and persistent storage**.

Ace Prep: Full-Stack Interview Simulator

April 2025 - June 2025

- Scaled a **React + Node.js AI powered SaaS** mock-interview platform to **200+** users, improving workflow reliability across onboarding, multi-step interview sessions, and transcript access.
- Reduced transcript retrieval latency by **35% (1.2s faster)** by redesigning the **PostgreSQL** schema, indexes, and query patterns to optimize high-read workloads.
- Increased release velocity by **20%** by implementing test-backed **CI/CD**, preventing regressions in session and transcript flows.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, TypeScript, JavaScript, SQL, Java
- **Backend & Distributed Systems:** REST APIs, Microservices, Distributed Systems, API Design, System Design
- **Cloud, Data & DevOps:** AWS, Docker, GitHub Actions CI/CD, PostgreSQL, MongoDB, DynamoDB, Firebase
- **Reliability & Operations:** Debugging, Monitoring, Logging, Unit Testing, Code Reviews, Technical Documentation
- **AI & Development Tools:** GitHub Copilot, Claude Code, LLM Integration, Prompt Engineering, RAG, AI Automation